Data curation and hyperparameter transfer enable competitive variant-effect prediction with a 1B genomic language model

TODO

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Abstract

Genomic language models can learn functional constraints directly from DNA sequence, but strong performance has often depended on specialized architectures, very large training budgets, or information derived from alignments and functional genomics. We developed MarinDNA to test how far a standard decoder-only Transformer can be taken through functional-region data curation, explicit mixture design, and systematic hyperparameter transfer. Hyperparameters calibrated on approximately 25M-parameter reference models identified the best observed learning rate at 255M, 476M, and 1B parameters, and the transferred recipe produced predictable loss scaling across eight models from 46M to 4B parameters. Continued training on a five-region mixture yielded a 1B model trained on 166B tokens that was statistically competitive with Evo 2 40B on the frozen Mendelian variant-effect benchmark. Relative to Evo 2 40B, this model used approximately 1,980-fold fewer reported training FLOPs and achieved approximately 2,330-fold higher as-deployed scoring throughput at each model's native context length, although alignment-based and functionally supervised models remained stronger overall. These results show that careful data and optimization choices can make a conventional, short-context genomic language model competitive at substantially lower training and deployment cost.

1. Introduction

Genomic language models (gLMs) learn statistical constraints directly from DNA sequence and can support variant-effect prediction, sequence design, and transfer learning (Benegas, Ye, et al., 2025). Early sequence-only models showed that such constraints can predict the effects of genome-wide variants without task-specific labels (Benegas et al., 2023). More recent work has expanded model scale, context length, biological coverage, and architectural specialization, but it remains unclear how much of the resulting performance depends on specialized architectures rather than the data and optimization recipe.

The distinction matters because the strongest genomic predictors often use information that is unavailable outside a small number of well-studied species. GPN-Star explicitly models whole-genome alignments and phylogeny (Ye et al., 2025), whereas AlphaGenome is supervised on thousands of human and mouse functional-genomics tracks (Avsec et al., 2026). Sequence-only models require neither input at inference time and can therefore be applied wherever an assembled genome is available. They do not yet replace alignment-based or functional-genomics-supervised models in humans, but they may provide a practical route to genome-wide constraint prediction in species that lack those resources. Human evaluation is consequently a stringent proxy for this broader goal, not direct evidence of cross-species transfer.

We treat training-data construction as a primary modeling variable. TraitGym found that a 152M-parameter promoter specialist was competitive with Evo 2 40B on promoter variants, while Evo 2 scaling produced smaller gains on distal regulatory variants than on several other consequence classes (Benegas, Eraslan, et al., 2025). Those observations motivate explicit control over functional-region sampling rather than uniform sampling from mostly neutral genomic background. MarinDNA begins with coding, upstream, and downstream regions available from standard gene annotations, then adds non-coding RNA and candidate cis-regulatory elements by projecting human annotations across species. The experiments vary which regions are present, their mixture weights, and the stage of training at which they are introduced.

To isolate those choices, we use a conventional decoder-only Transformer rather than a genomics-specific sequence operator. The model is a nucleotide-level Qwen3 causal language model with a 255-base context, chosen to model individual functional elements while keeping training and variant scoring inexpensive. An initial objective comparison favored causal language modeling for subsequent development, but was not a definitive matched-compute comparison. Long-range regulatory modeling and bidirectional adaptation remain outside the scope of this study.

We evaluate frozen checkpoints on clinically curated Mendelian variants and saturation-genome-editing measurements. Each model is assessed through two complementary readouts: a zero-shot reference-versus-alternate sequence log-likelihood ratio and a chromosome-disjoint linear probe on paired allele embeddings. The likelihood score tests whether the training objective itself ranks deleterious alleles; the probe tests whether variant-relevant information is present in the learned representation even when it is not exposed by sequence likelihood. These evaluations measure human functional constraint and pathogenicity, not gene-expression changes or organism-level fitness.

Evo 2 40B is the principal sequence-only baseline because it is broadly evaluated across genomic regions and was trained at a substantially larger scale: 40B parameters, approximately 9.3T DNA tokens, and 2.25e24 reported training FLOPs (Brixi et al., 2026). Against that baseline, we ask whether a simpler and much smaller model can be made competitive through data curation and transferable optimization. We report three connected findings: optimizer settings calibrated at small scale transfer across model size, token horizon, and batch size; validation loss follows a predictable parameter-scaling trend but zero-shot missense performance does not improve monotonically; and a staged five-region training mixture yields a 1B-parameter model that is competitive with Evo 2 40B on the frozen Mendelian benchmark while requiring far less training compute and delivering higher as-deployed inference throughput.

2. Results

2.1. Early mixture experiments

Across the study, training examples came from annotation-derived coding, upstream, and downstream regions and from alignment-projected non-coding RNA and candidate cis-regulatory-element annotations. Figure 1 summarizes the provenance and scale of these datasets; individual experiments used the versions and mixtures specified below and in the supplementary provenance table.

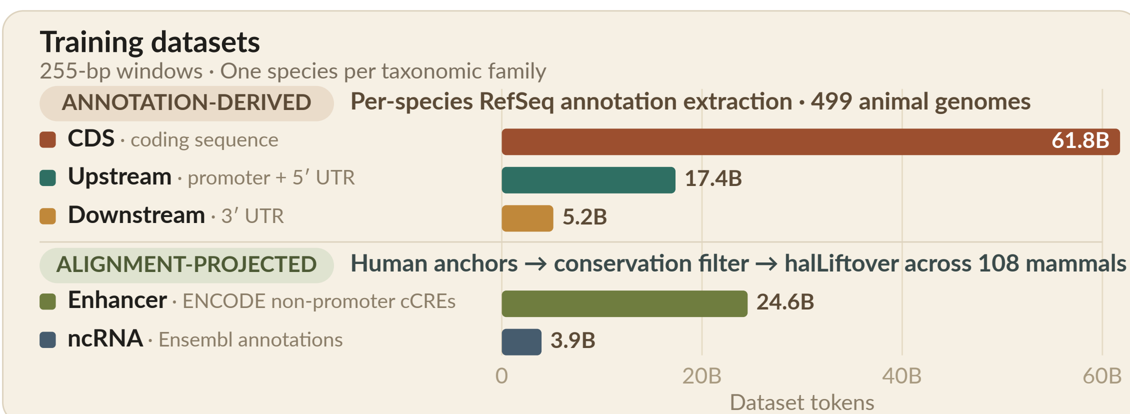

Figure 1 **Training-dataset provenance and scale.**

We first trained a 1.7B upstream-region specialist, trying to replicate the success of GPN-Promoter (experiment #21).[1] Although we used reasonable defaults rather than the systematic hyperparameter-transfer recipe developed later, performance was broadly comparable to Evo 2 40B. We saw a similar pattern when training a CDS specialist (experiment #27). Overall, however, GPN-Star remained stronger.

[1]The upstream and CDS datasets used in these early experiments were earlier versions of those summarized in Figure 1: broadly comparable, but not identical. These models also used a 512-bp context without a BOS token, roughly twice the 255-bp context adopted for the later recipe.

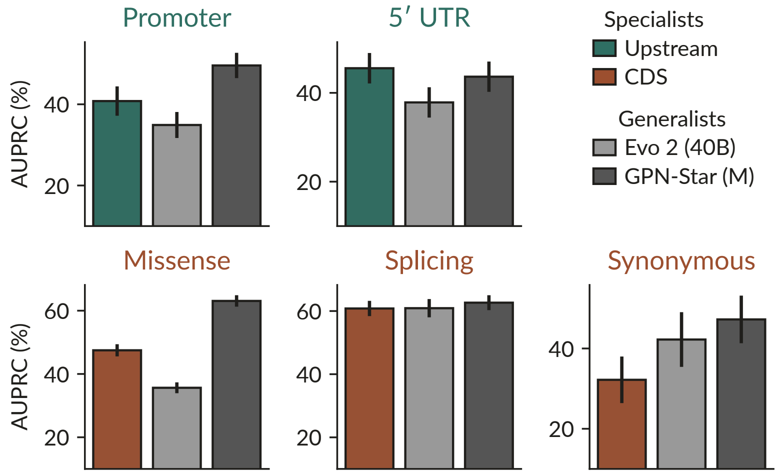

Figure 2 **Region-matched zero-shot Mendelian VEP: specialists versus generalists.** Error bars denote SE.

After testing upstream and CDS specialists independently, the next experiment asked whether one model could retain both capabilities (experiment #13). Sampling in proportion to dataset size—10% upstream and 90% CDS—is the naive default when the two datasets are simply pooled without reweighting. In practice, it behaved similarly to CDS-only training. Equal 50/50 upstream/CDS sampling produced balanced performance across both regions. This made explicit mixture control a central axis of investigation. Even the 50/50 mixture may not be optimal: regions can differ both in size and in the density of learnable biological signal.

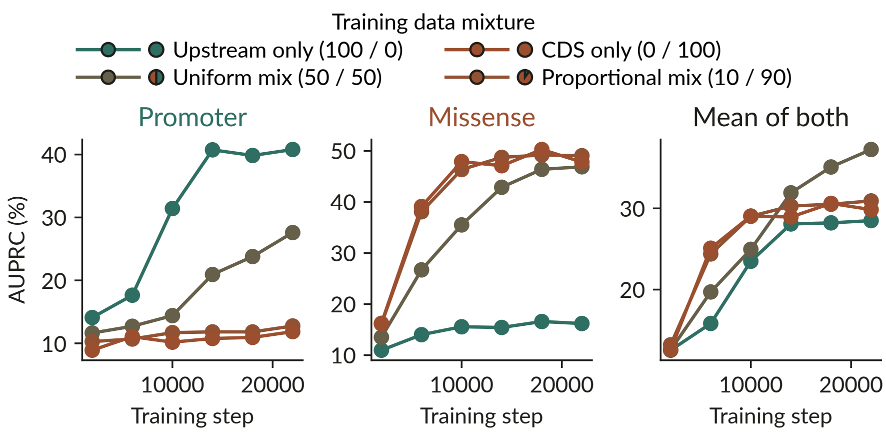

Figure 3 **Upstream/CDS training-mixture comparison.** Zero-shot results; the right panel is the unweighted mean of the promoter and missense AUPRC values.

2.2. Hyperparameter transfer

Our first attempt to scale manually lowered the learning rate as the model grew from 0.6B to 1.7B and 4B parameters (experiment #57). The larger models did not improve over the 0.6B model, and the early 4B run became unstable. That failure made systematic hyperparameter transfer a prerequisite: without a trustworthy optimization recipe, a model-size comparison would confound scale with tuning quality.

The annotation-derived DNA pool available at the time contained ~85B tokens. Figure 4 shows its proportional CDS, upstream, and downstream composition.

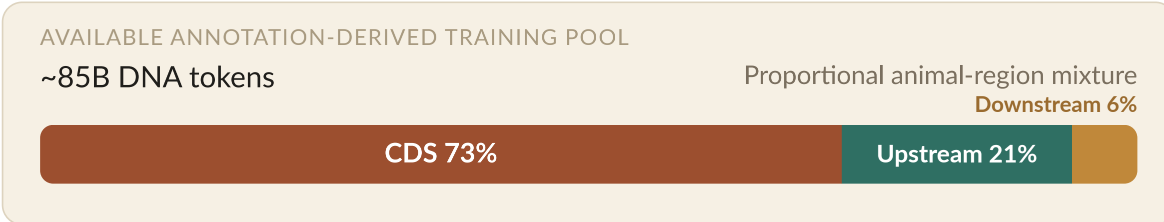

Figure 4 **Available annotation-derived training pool.**

That pool is large by genomics standards but small relative to modern accelerator-era training corpora. This makes the project data-constrained in principle even though our practical constraints are messier. We train on preemptible Google TPU Research Cloud resources, do not have consistent access to slices much larger than roughly 32 H100s worth of peak FLOPs, and want the recipe to remain reproducible at academic compute scale. O(100B) tokens therefore lands in an awkward middle ground where compute-constrained methods are still relevant, even though modest epoching is possible and likely breaks their assumptions at some unknown rate.

We started with hyperparameter transfer for that reason. If a proven data-constrained transfer framework existed, we would use it. We do not know of one, so we followed the same basic pattern as Delphi, fitting a small reference sweep with the Vizier Bayesian optimization framework and then scaling the result using a Complete(d)-inspired AdamH heuristic.[2]

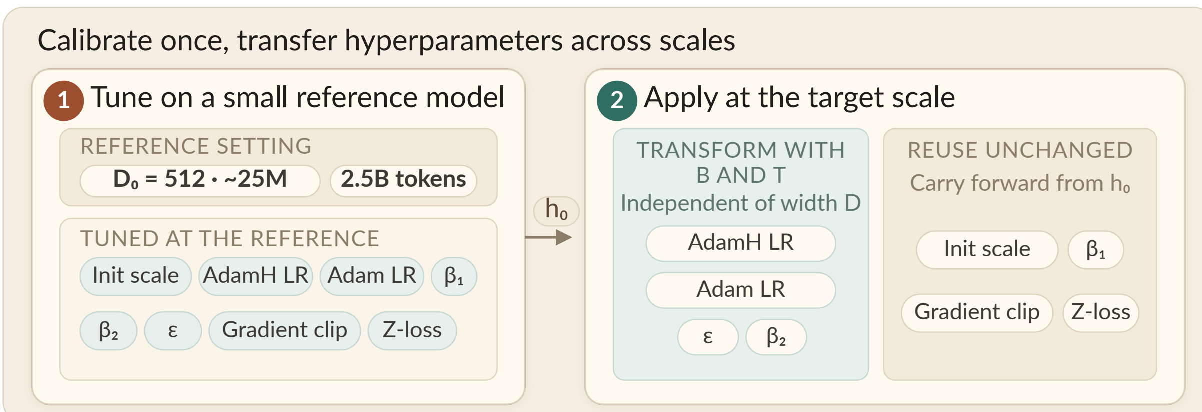

Figure 5 **Hyperparameter calibration and transfer.**

Figure 5 separates reference calibration from target application. Two heuristics sit behind that workflow and are fixed before tuning: an inherited rule maps hidden width D to model geometry,[3] and a second rule maps reference optimizer settings to a new batch size B and token horizon T. The reference sweep tunes initialization scale, the two learning rates, β_1, β_2, ε, gradient clipping, and z-loss. Here, tuning a learning rate means tuning its peak value under a fixed fractional schedule; the target run reuses that schedule shape, so warmup and decay scale with the run length rather than keeping fixed absolute step counts.[4] At the target, the two learning rates, β_2, and ε are transformed with B and T, while initialization scale, β_1, gradient clipping, and z-loss are reused unchanged.[5]

The reference sweep used ~25M-parameter models trained for 2.5B tokens with a 16k-token batch, or roughly 4e17 FLOPs per run. We then validated the transferred hyperparameters across 255M–1B-parameter models, with 4x as many tokens, 1/4x the batch size, and roughly 170x the FLOPs per run. The first test was whether the learning-

[2]Complete(d) refers to the compute-constrained hyperparameter-transfer framework described in Complete(d): Data-Optimizing Hyperparameter Transfer.

[3]The DNA experiment inherits its geometry rule from Marin's text-model scaling heuristic rather than fitting it on DNA. For hidden width D, it sets $\text{layers} = \text{round}(D/(55 + 4\cdot\log_2 D))$, $\text{MLP width} = 4D$, and both attention-head and KV-head counts to $D/128$; the sweep widths are selected manually. See the commit-pinned geometry rule and model builder.

[4]Both learning rates use 10% linear warmup, remain at their peak through 80% of training, and then decay linearly to zero over the final 20%. These are fractions of the target run's total steps. See the experiment configuration and scheduler implementation.

[5]Relative to the reference batch and token horizon (B_0, T_0), the fixed heuristic uses $\text{AdamH LR} \propto \sqrt{(B/B_0)}\cdot(T_0/T)^{0.3}$, $\text{Adam LR} \propto \sqrt{(r/r_0)}$, $\varepsilon \propto \sqrt{(r_0/r)}$, and $\beta_2 = \text{clip}(\beta_{2,0}{}^{(B/B_0)})$, where $r/r_0 = (B\cdot T_0)/(B_0\cdot T)$ and configured bounds still apply. The 0.3 exponent is inherited from Marin's text recipe rather than fitted on the DNA sweep; the Complete(d) paper proposes a 0.5 token-horizon exponent, while a later Marin text sweep estimated ~0.28. See the commit-pinned implementation.

rate prediction survived that regime. Figure 6 shows that the transferred prediction lands exactly on the best observed learning-rate setting at all three validation scales, outperforming both the unchanged reference optimum and every other target-scale sweep setting; the less sensitive optimizer hyperparameters are shown separately in Figure 7.

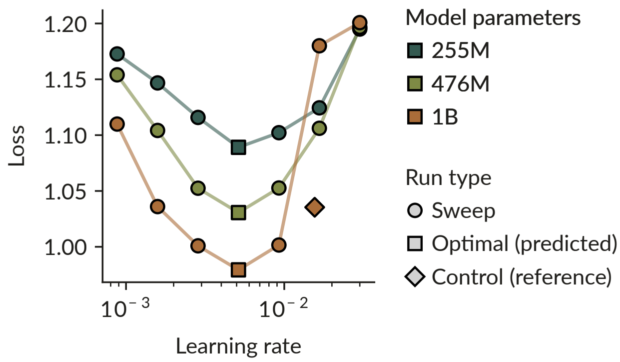

Figure 6 **Learning-rate transfer across model scales.** Results across the 255M, 476M, and 1B validation scales. The control run type indicates final loss from the optimal configuration found in the initial smaller-scale reference sweep. The predicted, optimal LR results in a better loss than both this control and all other configurations at the same scale (sweep run type), for all model sizes.

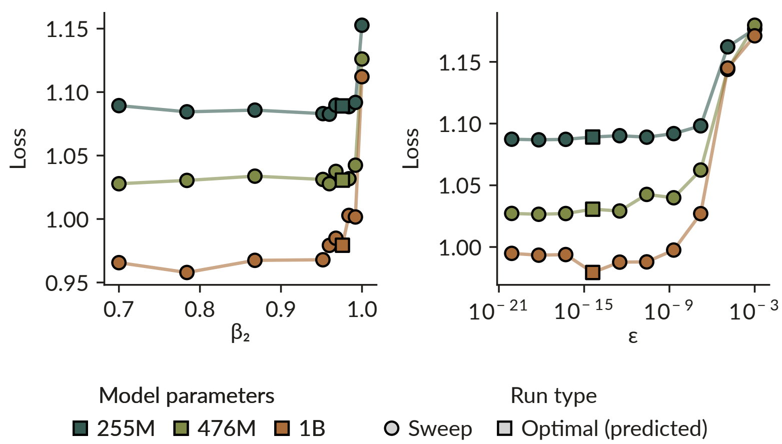

Figure 7 **Adam β_2 and ε transfer.** Results across the same scales as Figure 6.

That validation is a fairly unforgiving test. If the transferred learning rate were merely close by accident, it would be surprising for it to land correctly across all three validation scales, but the prediction remains well centered at each one. For DNA, that is a pretty cool result. Prior biology foundation-model work has used μP-style transfer, but we are not aware of a DNA result showing that a more inclusive framework like Complete(d) works across token horizon and batch size, which are the axes we keep leaning on later in ad hoc runs across epochs. The same is mostly true for the other optimizer hyperparameters too, although Adam β_2 shows some signs of being a bit aggressive at the largest scale. Supplementary Figure S1 makes the same point across CDS, upstream, and downstream sequence, with no qualitative difference in transfer behavior across region types. That gives us enough confidence that the following parameter-scaling runs are at least close to optimally configured.

[6]Here, "validation loss" is best understood as a training-loss-like monitoring statistic computed on a fixed set of human training sequences, rather than a conventional estimate on held-out data. We have not yet found a satisfactory way to construct clean held-out genomic splits: genomes are phylogenetically correlated, and identifying orthologous non-coding regions by sequence alignment is difficult. See issue #8 for split experiments and a broader discussion of why raw perplexity may not reliably track VEP performance. The meaning of lowercase differs between training and validation: during training it marks repetitive bases, whereas in these validation

2.3. Parameter scaling

Before asking whether better validation loss[6] translates into better VEP performance, we first needed to check whether validation loss scaled the way it should. The parameter sweep uses the same training recipe at each model size, with all hyperparameters set by the transfer heuristic above, and then asks whether the resulting losses fit a Kaplan-style scaling law well (they do).[7] Despite this being a simple experiment conceptually, actually getting there took months — fitting the hyperparameter transfer heuristic, running the validation experiments, and training the 4B model, which alone took about three weeks to finish. The final sweep spans 8 model sizes from 46M to 4B parameters, each trained on ~84B tokens, for ~4.3e21 FLOPs across the sweep. That puts it on par with canonical scaling-law studies in language modeling, e.g. its ~2.1e21 FLOP 4B run matches the compute Hugging Face used at that exact model scale in their data-constrained scaling work.[8]

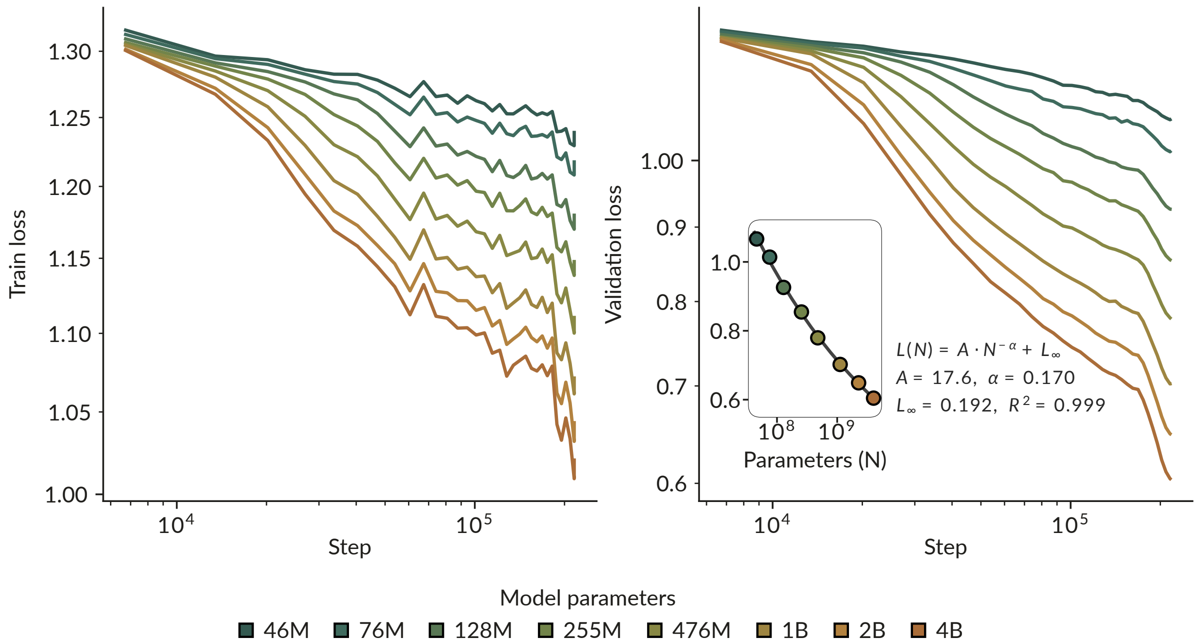

Figure 8 **Loss scaling across model sizes.** Training and validation loss for eight models from 46M to 4B parameters, each trained on ~84B tokens; the inset shows a Kaplan power-law fit to final validation loss.

The result is about as tidy as we could hope for. Training is stable at every scale, and both training and validation loss decrease monotonically and predictably, as shown in Figure 8. We use WSD learning-rate schedules with 10% warmup and 20% decay, which causes the visible drop in both losses over the final 20% of tokens. Most importantly, the sweep gives a high-quality Kaplan scaling-law fit (R2=0.999), which makes the next question much better posed. Does lower validation loss actually correlate with better downstream VEP performance?

2.4. Downstream performance

The final sweep shows a mostly consistent relationship between parameter count and downstream VEP performance. When zero-shot LLR and linear probes are evaluated on identical variants, performance improves with scale for most variant types. The clearest exception is Mendelian missense: zero-shot LLR peaks at 128M parame-

sets it marks non-conserved bases; in both cases lowercase positions receive 1% of the standard loss weight. In validation, this acts as a heuristic that emphasizes conserved sequence, but we did not independently design or validate it as a model-selection objective. Because this statistic informed the Vizier reference sweep and the choice of the 1B model, those decisions may not be optimal under a different validation objective. We therefore use validation loss descriptively for like-for-like comparisons and trend analysis, not as an unbiased estimate of performance on unseen sequence; the biological conclusions rely primarily on downstream VEP evaluation.

[7] This follows the empirical scaling-law setup from Kaplan et al., where model loss is fit as a predictable function of model size, data, and compute.

[8] See Fig. 4 of Muennighoff et al., "Scaling Data-Constrained Language Models" (NeurIPS 2023).

[9] Non-monotonic likelihood-based zero-shot variant-effect performance with increasing model scale has previously been observed in other settings. Gordon et al. report that ESM-2 performance on protein deep-mutational-scanning benchmarks degrades beyond

ters and then deteriorates, even as linear-probe performance continues to improve.[9] This is not a general failure on missense variants—the SGE missense benchmark improves with scale under both scoring protocols.

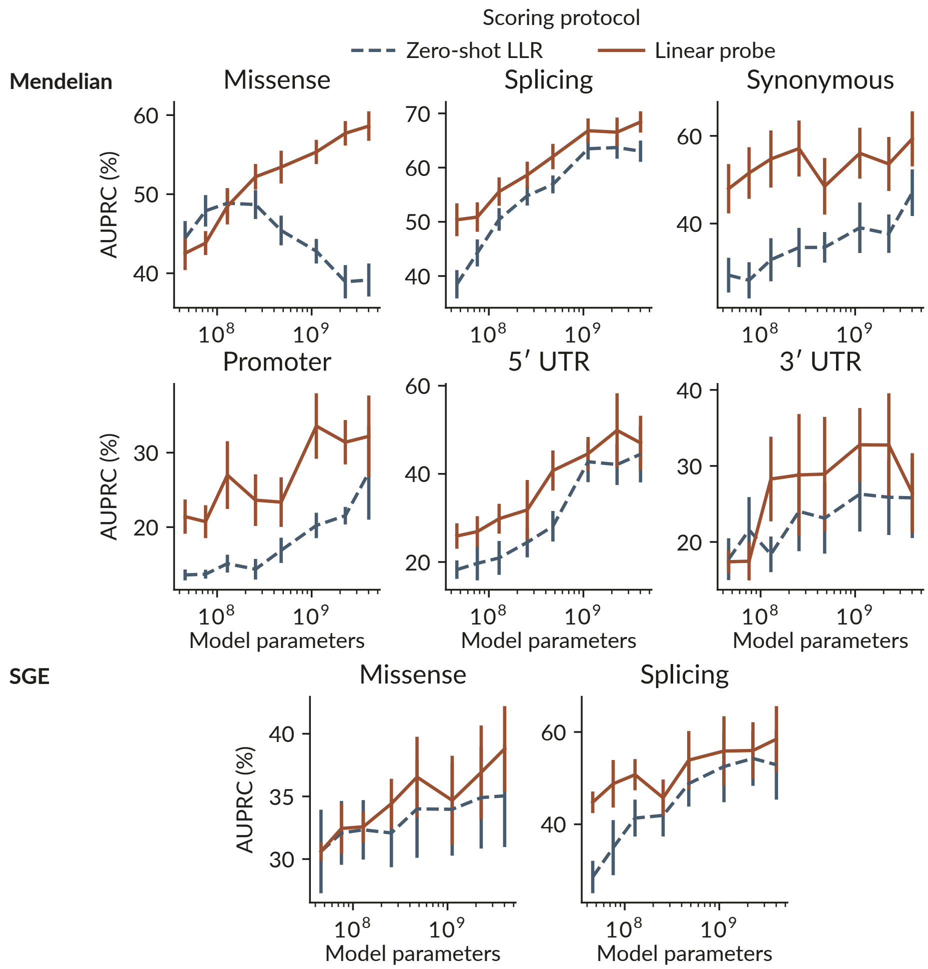

Figure 9 **VEP performance across model scale.** Zero-shot LLR and a linear probe are compared on identical variants. Performance is measured as chromosome-weighted AUPRC; error bars denote SE.

Plotting the same results against matched-region validation log-likelihood gives the same picture. For all other combinations of variant type and scoring protocol, better validation log-likelihood is associated with better downstream performance. Zero-shot Mendelian missense again points in the opposite direction: performance declines even as validation log-likelihood improves.

an intermediate model size and show that performance depends on the likelihood assigned to the wild-type sequence. Pugh et al. similarly report plateaus or regressions for larger protein language models under standard likelihood-based scoring. These studies concern protein models and different evaluation regimes; we do not know whether our Mendelian missense result has the same cause.

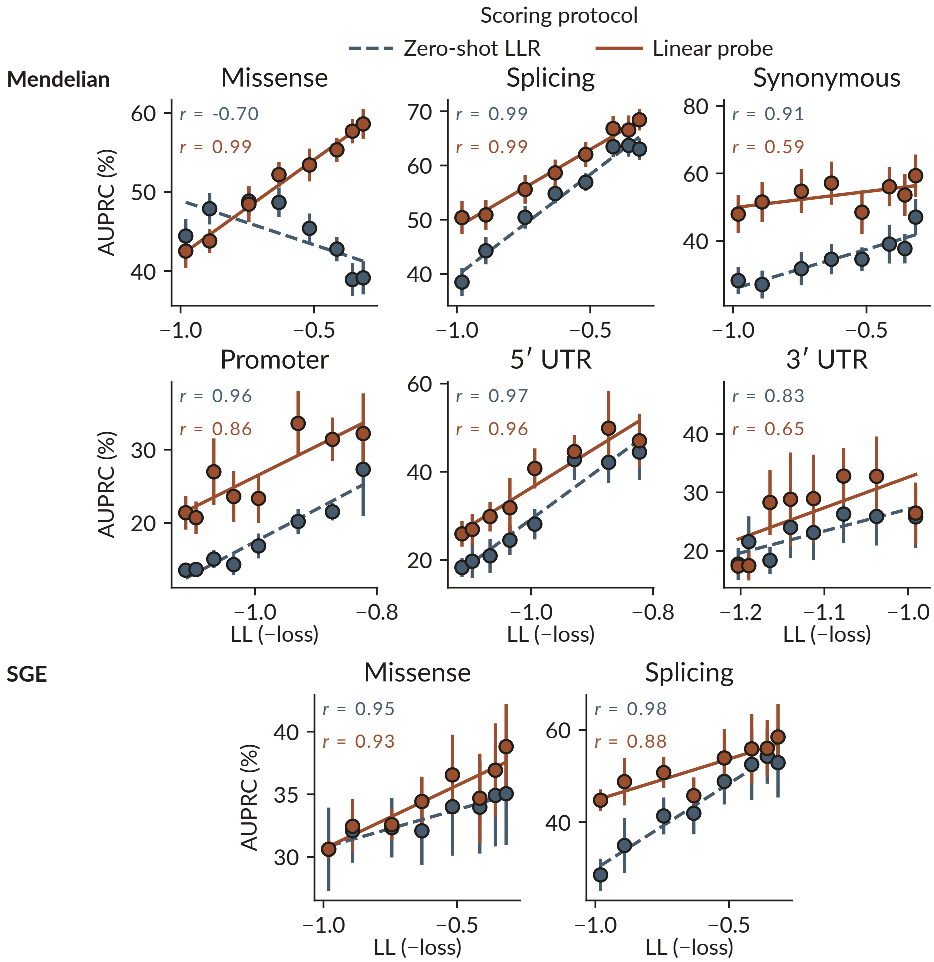

Figure 10 **VEP performance versus validation log-likelihood.** Matched-region LL (shown as −loss) across the eight parameter-scaling endpoints. Performance is measured as chromosome-weighted AUPRC. Lines are least-squares fits, and r denotes Pearson correlation; error bars denote SE.

This divergence is not unique to MarinDNA. On the same Mendelian missense benchmark, Evo 2 also shows improving linear-probe performance alongside declining zero-shot LLR performance as model size increases. For now, this should be interpreted as a recurring pattern for Mendelian missense across these two model families—not as evidence that zero-shot readouts generally deteriorate with scale. It also cautions against using zero-shot LLR alone to judge whether scaling has improved the learned representations for this task.

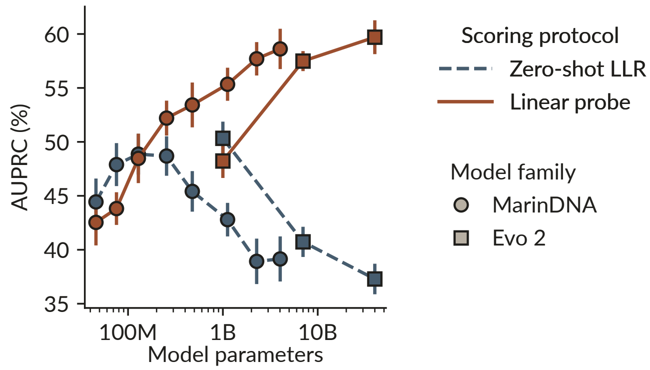

Figure 11 **Missense readouts across model scale.** MarinDNA and Evo 2 are compared using zero-shot LLR and a linear probe on identical Mendelian variants. Performance is measured as chromosome-weighted AUPRC; error bars denote SE.

2.5. Later mixture experiments

At this point we move away from theoretically grounded, compute-constrained methods. The later experiments still rely on the transfer heuristics above, since we need learning rates and other hyperparameters for runs with very different token horizons. But the actual optimization problem becomes much more ad hoc — we start changing mixture constituents, epoch them freely, and see whether in-flight changes can compensate for observed performance gaps.

We standardized these experiments on a 1B-parameter model.[10]

Our starting point was a uniform mixture of CDS, upstream, and downstream sequence. These were the three region datasets we could initially construct consistently across many species from standard genome annotations. The scaling study had instead sampled these regions in proportion to dataset size. Echoing the earlier mixture experiments, that recipe produced good CDS performance but left a large gap on the less abundant upstream tasks. We therefore switched to uniform weighting so that each functional region received meaningful exposure.

We use the internal labels m5.1, m1.3, and m3.3 for the three model-mixture lineages compared below. In this naming scheme, m denotes a mixture lineage, the leading number identifies the mixture strategy, and the suffix identifies a continuation within that lineage. m5.1 is the staged lineage: we first trained its 1B model for ~104B tokens on the uniform three-region mixture. Alignment projection then made it possible to turn human ncRNA and enhancer annotations into comparable multi-species training datasets. Once those data became available, we added them to form a uniform five-region mixture and continued training the same model for another ~62B tokens. We compare this staged history with two controls, m1.3 and m3.3, trained on five-region mixtures from the beginning.

[10] At the time, 1B had reached a good level of zero-shot performance under our then-current evaluation. We had not established it as the optimal model size. The later linear-probe results change this judgment most: they continued to improve with scale even where zero-shot Mendelian missense did not, making a larger model a more compelling choice in retrospect.

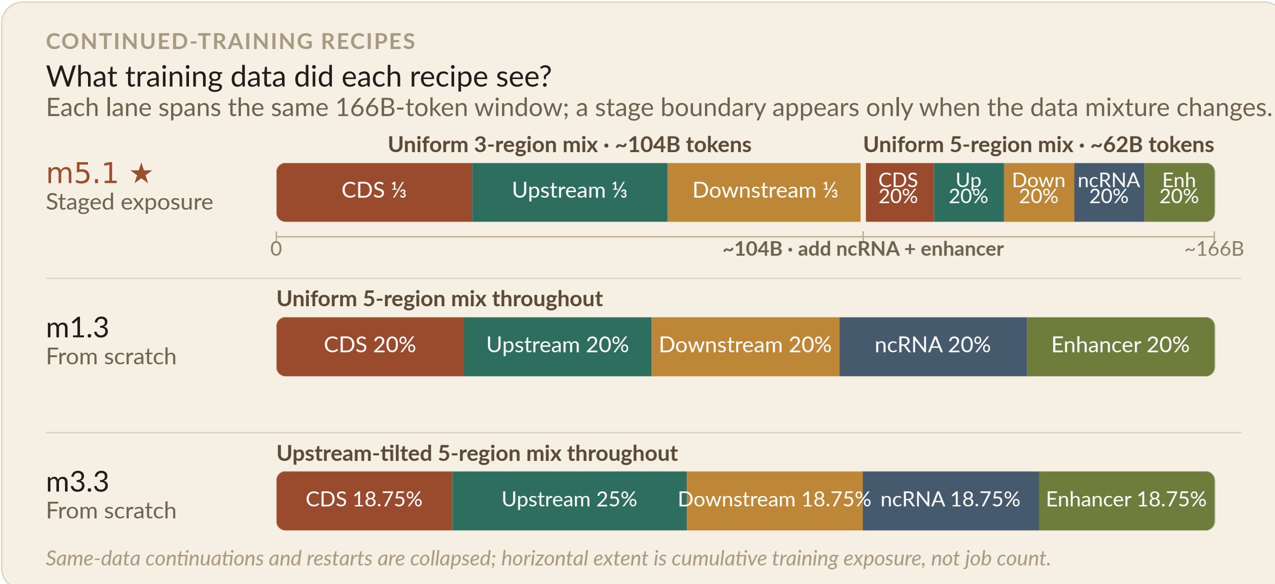

Figure 12 **Training-data exposure histories.** Three recipes are compared. m5.1 trains for approximately 104B tokens on a uniform three-region mixture before adding ncRNA and enhancer data for approximately 62B tokens. The de novo m1.3 and m3.3 controls keep fixed five-region mixtures over the same displayed token horizon; m3.3 gives upstream sequence 25% weight and each other region 18.75%.

For m5.1, the first evaluated checkpoint after adding ncRNA and enhancer data shows gains in variant-effect performance in the corresponding ncRNA and distal subsets. Across all eight subsets, m5.1 ultimately finishes with the highest macro-average AUPRC under both zero-shot LLR and the linear probe, although the linear-probe trajectories are visibly noisier.[11] Its advantage is broad but not universal: the lineages trained on five regions from the beginning retain stronger endpoints for some distal and ncRNA subsets. m5.1's strong endpoint raises the possibility that exposure order matters: learning first from the three-region mixture and introducing ncRNA and enhancer data later may be more effective than training on all five regions from the beginning, though this remains uncertain and requires further investigation.

The corresponding linear-probe trajectories appear in Supplementary Figure S2.

[11] The curves shown here use a separate probe trained within each variant subset. In a separate 255M analysis on one held-out chromosome, training one probe across all subsets improved the AUPRC point estimate on several data-starved subsets, including ncRNA and distal, while hurting stronger or more specialized subsets. This makes limited labeled data one plausible contributor to the noise, but that analysis did not directly test the checkpoint-to-checkpoint variability in these 1B lineage curves.

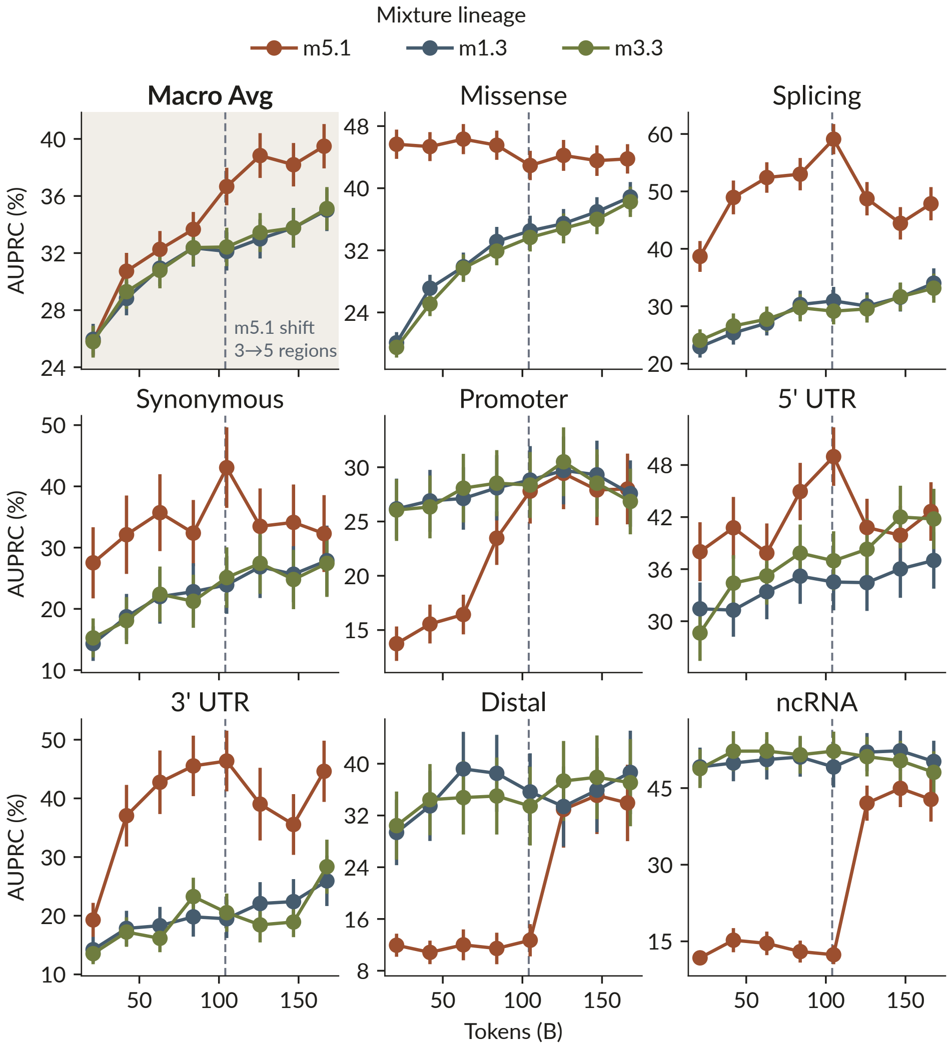

Figure 13 **Zero-shot mixture-lineage trajectories.** Mendelian AUPRC versus training tokens for three model-mixture lineages. Error bars denote SE.

2.6. Leaderboard scores

The result of the previous mixture experiments is m5.1, a 1B GPT-style model evaluated alongside other models on our live Mendelian VEP leaderboard, where we continue to add experimental runs and baselines. In the zero-shot snapshot shown here, m5.1 comes out slightly ahead of Evo 2 40B. Its advantage is considerably larger under linear probing.

m5.1 was trained on 166B tokens (~1.1e21 FLOPs), compared with 9.3T tokens (~2.25e24 FLOPs) for Evo 2 40B. At their native context lengths on the same GH200, m5.1 scores one million variants in about 41 minutes, compared with roughly 66 days for Evo 2 40B—a roughly 2,330× throughput advantage.[12]

[12] This benchmark measures steady-state scoring with forward and reverse-complement passes and embeddings enabled. m5.1 uses a 256-token context, while Evo 2 40B uses an 8,192-token context, so this measures as-deployed throughput rather than same-context or per-token efficiency. See issue #354 for the full methodology and results.

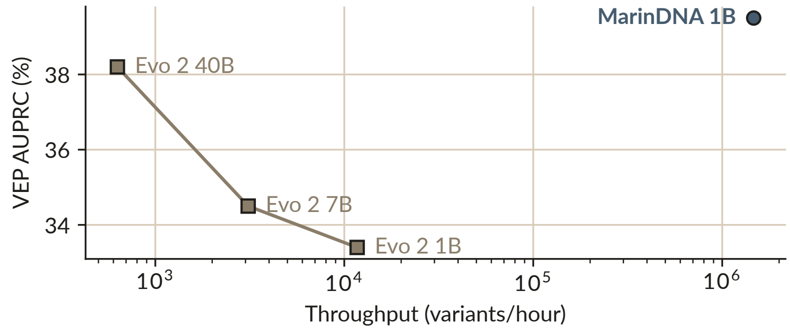

Figure 14 **Variant-effect performance versus as-deployed inference throughput.** Zero-shot Mendelian macro-average AUPRC versus variants scored per hour on one GH200. Each model is evaluated at its native context length; throughput includes forward- and reverse-complement scoring and embedding output.

Most notably, m5.1 closes Evo 2's main gap on distal variants, outperforming Evo 2 40B there under both readouts. The improvement is not uniform, however: Evo 2 40B remains ahead on splicing under both readouts, as well as on promoter and synonymous variants in the zero-shot evaluation and missense variants under linear probing.

The current leaderboard also suggests there is considerable headroom: although m5.1 leads on Macro Avg among MarinDNA models, another MarinDNA run has a higher point estimate in seven of the eight displayed subsets, with m5.1 leading only on ncRNA. Several of these winners are region specialists, suggesting that further mixture refinement could recover more of their complementary strengths.

On the broader zero-shot leaderboard, m5.1 remains slightly behind AlphaGenome and substantially behind GPN-Star.[13] These are different model families: AlphaGenome learns from functional-genomics supervision, while GPN-Star uses whole-genome alignments. Further improvements to the alignment-free recipe may narrow the gap to GPN-Star, but it is not clear that they will eliminate it. Some of the remaining difference may reflect information that an alignment-free model cannot recover from unaligned sequence alone (research question #397).

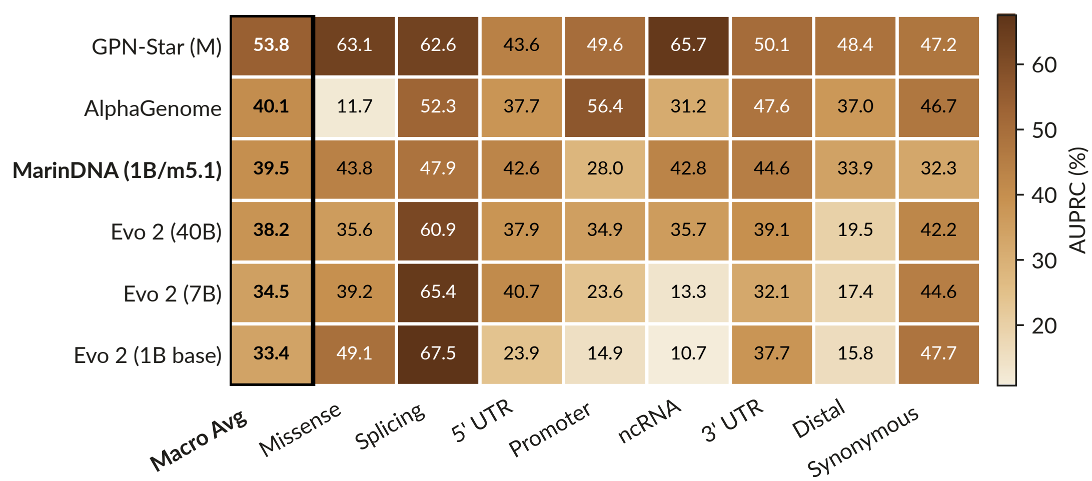

Figure 15 **Zero-shot Mendelian leaderboard.**

[13]Figure 15 uses LLR for MarinDNA and Evo 2, calibrated LLR (cLLR) for GPN-Star, and the maximum L2 REF/ALT prediction score for AlphaGenome.

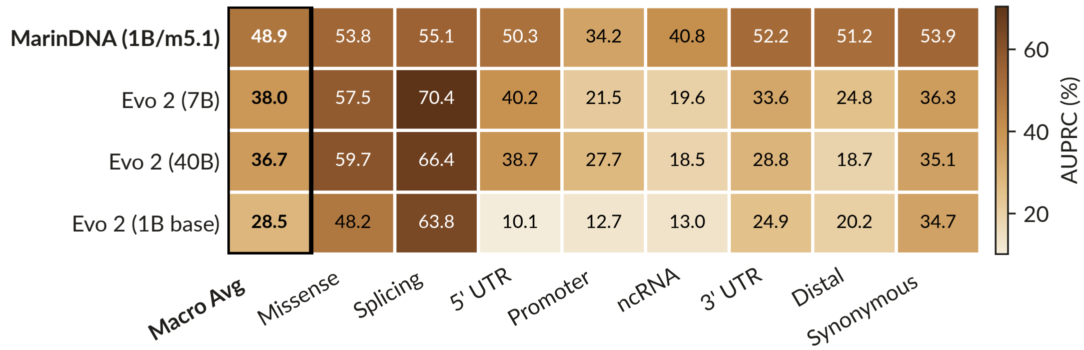

Figure 16 **Linear-probe Mendelian leaderboard.** AUPRC is a sample-size-weighted average across chromosomes and is not directly comparable with Figure 15. GPN-Star and AlphaGenome are absent because no compatible probe results are available here, not because of their performance.

3. Discussion

MarinDNA shows that a conventional decoder-only Transformer can become a strong genomic sequence model when data construction and optimization are treated as primary modeling choices. The final 1B model establishes the attainable performance and efficiency, while the mixture and hyperparameter-transfer experiments explain how that result was reached.

The data-mixture results indicate that proportional sampling is not a neutral default when genomic region datasets differ substantially in size. In the upstream/CDS experiment, proportional pooling behaved similarly to CDS-only training, whereas equal sampling retained useful performance across both region types. The later m5.1 lineage extended this principle by introducing alignment-projected ncRNA and enhancer data and finished with the strongest macro-average endpoint among the three frozen mixture lineages. This result supports explicit control of region exposure, but it does not establish that staged exposure is intrinsically better than de novo five-region training because the lineages are not a controlled curriculum ablation.

The hyperparameter experiments provide complementary evidence that optimization settings can be transferred across model size, batch size, and token horizon in this regime. The transferred learning rate was the best observed tested value at each of three target scales, and the resulting parameter ladder trained stably through 4B parameters. The transfer rule nevertheless inherits assumptions from a text-model recipe, including its token-horizon exponent and model-geometry heuristic, and the present experiments do not establish that those choices are globally optimal for DNA.

On the frozen Mendelian evaluation, the final MarinDNA model is statistically competitive with Evo 2 40B rather than demonstrably superior to it. Point estimates differ by consequence class and readout: MarinDNA closes much of the distal-variant gap, while Evo 2 retains advantages on several coding and splicing subsets. GPN-Star and AlphaGenome remain stronger in the broader zero-shot comparison, but they use alignment-derived information and functional-genomics supervision, respectively, and therefore bound rather than directly match the alignment-free setting studied here.

The compute and throughput comparisons require similar care. The training-FLOP ratio uses reported total training compute, whereas the throughput measurement compares steady-state scoring on the same GH200 at each model's native context length and includes forward- and reverse-complement passes plus embedding output. It therefore measures the cost of deploying the evaluated models for this workload, not architecture-only efficiency at matched context length or matched tokens processed.

Several limitations constrain the biological interpretation. The models use a short 255-base context chosen for functional-element evaluation, so these experiments do not test long-range regulatory interactions or genome-scale sequence generation. Human variant-effect data provide the strongest available evaluation, but performance in humans does not by itself establish transfer to non-model organisms, which is an important motivation for alignment-free models. The weighted validation loss is computed on fixed human sequences related to the training distribution rather than a clean phylogenetically independent holdout, so it is useful for controlled comparisons but not an unbiased estimate of genomic generalization. Finally, the zero-shot Mendelian missense regression with scale shows that likelihood-based readouts can diverge from information retained in frozen representations.

Neither scaling nor optimization appears exhausted: linear-probe performance continued to improve through the largest evaluated model, and the loss-scaling fit showed no clear saturation over the tested range.

4. Methods

4.1. Study design and frozen analysis snapshot

This manuscript reports the model, evaluation, and figure snapshot used for the published MarinDNA article rather than a continuously updated leaderboard. The mechanical manuscript baseline and its 20 SVG assets were fixed at MarinDNA commit 3b608d39b41c2330636ec647dbb25d26b0895187, whose content and figures were converted from project commit d8c4803cbbbffafb24890cd0c75134d78368d55c. No new training run or evaluation dataset was introduced for the editorial conversion. The live leaderboard is treated as a separate, evolving resource.

4.2. Training data

Training examples were drawn from functional regions across multiple animal genomes. The initial datasets contained coding sequences (CDS), sequence upstream of annotated genes, and sequence downstream of CDS ends. The downstream category denotes the 256 bases immediately after an annotated CDS end rather than annotated 3′ UTR intervals. Later training mixtures added functional ncRNA regions and ENCODE V4 non-promoter candidate cis-regulatory elements, with human annotations projected to other species by sequence alignment. All genomic coordinates inside MarinDNA use 0-based, half-open intervals, with conversion to external coordinate conventions only at tool boundaries. The later recipe represents each example with 255 DNA bases and a beginning-of-sequence token, giving a 256-token model input. Soft-masked repetitive bases receive 1% of the standard training-loss weight. Training-data revisions and per-region token counts for every retained experiment will be enumerated in the supplementary provenance table.

4.3. Model and training objective

MarinDNA uses the Qwen3 decoder-only Transformer architecture with a causal next-token objective and nucleotide-level DNA tokenization. The architecture is intentionally conventional: the experiments vary training data, optimization, and parameter scale without introducing a genomics-specific sequence operator. The final m5.1 model has approximately 1B parameters and a 255-base context. It was first trained for approximately 104B tokens on a uniform mixture of CDS, upstream, and downstream data, then continued for approximately 62B tokens after ncRNA and enhancer data were added to form a uniform five-region mixture. The frozen internal checkpoint is mix-v0.9-p1B-i24-exp135-m5.1-step-59158 at GCS path gs: An immutable public model revision will be recorded in Section 5 before preprint release.

4.4. Hyperparameter calibration and transfer

A Bayesian reference sweep using Vizier trained approximately 25M-parameter models for 2.5B tokens with a 16k-token batch. The sweep tuned initialization scale, AdamH and Adam peak learning rates, β_1, β_2, ε, gradient clipping, and z-loss against the weighted validation-loss statistic described below. A fixed geometry rule maps hidden width to the number of layers, MLP width, attention heads, and key-value heads. A second fixed rule transfers the reference optimizer settings to a target batch size and token horizon. Relative to reference batch size B_0 and horizon T_0, the AdamH learning rate is multiplied by $\sqrt{(B/B_0)}(T_0/T)$^0.3. The Adam learning rate is multiplied by $\sqrt{(r/r_0)}$, ε by $\sqrt{(r_0/r)}$, and β_2 is exponentiated by B/B_0 subject to configured bounds, where $r/r_0 = (B\ T_0)/(B_0\ T)$. Initialization scale, β_1, gradient clipping, and z-loss are reused without transformation. Both learning rates use linear warmup over the first 10% of steps, remain at their peak through 80% of training, and decay linearly to zero over the final 20%. Target-scale validation used 255M-, 476M-, and 1B-parameter models with four times the token horizon and one quarter of the reference batch size. “Best observed” refers to the lowest final validation loss among the tested target-scale settings and does not imply a global optimum.

4.5. Parameter-scaling experiment and validation loss

The parameter ladder contains models of approximately 46M, 76M, 128M, 255M, 476M, 1B, 2B, and 4B parameters. Each model was trained for approximately 84.77B tokens with the transferred optimization recipe. The final validation losses were fit with a Kaplan-style power law as a function of parameter count. Training and validation losses weight soft-masked or non-conserved lowercase bases at 1% of the standard token weight. The validation statistic is computed on fixed human sequences related to the training distribution and is used for controlled comparisons and model selection. It is not interpreted as an unbiased estimate on a phylogenetically independent held-out genome set.

4.6. Variant-effect datasets

The Mendelian benchmark compares pathogenic variants with non-rare gnomAD controls matched within consequence class. Relative to the TraitGym design, the minimum control allele frequency is 0.1%, the consequence set includes missense and splicing variants, and matching includes potential confounders such as transcription-start-site distance and exon distance where applicable. The frozen benchmark revision is 4aed58e50c5dea0b878a665007af2ef9e5108e9f. The saturation-genome-editing benchmark is the v3 labeled build at 225d3d1ea32a4af547891b13c33b5e92a5aae849. It retains variants labeled abnormal or normal by ClinGen/ExCALIBR-calibrated assay thresholds and groups them into missense and splicing subsets. Variant contexts are extracted from the Ensembl release 115 GRCh38 soft-masked primary assembly. The evaluation pipeline uses the development split for model iteration; the chromosome-disjoint test split remains reserved for the final locked evaluation.

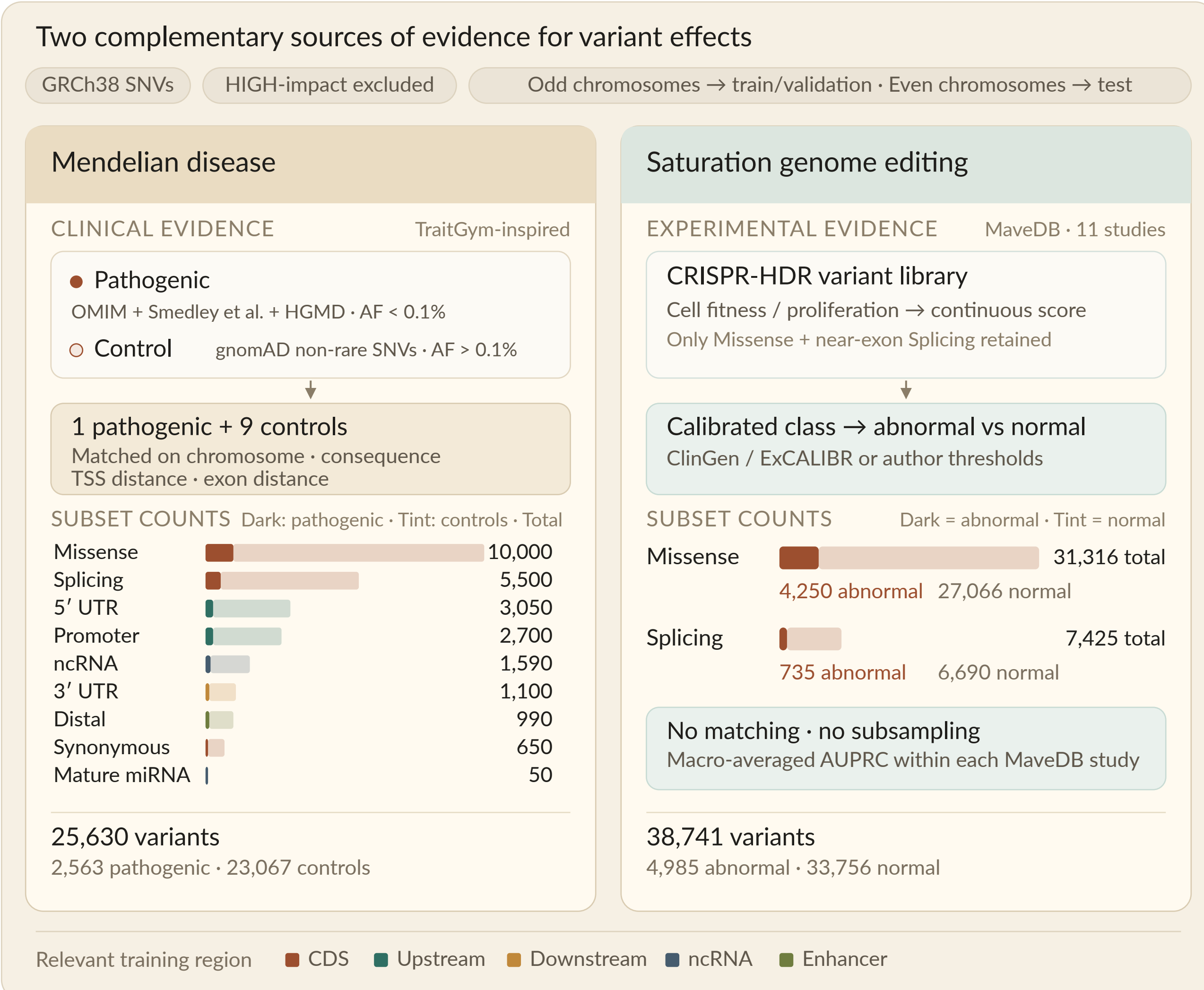

Figure 17 **Variant-effect prediction benchmarks.** Counts denote the frozen analysis snapshot described in the text.

The two benchmarks and their analysis subsets are summarized in Figure 17.

4.7. Zero-shot likelihood scoring

For each allele, the causal model scores the variant-centered sequence on the forward strand and its reverse complement. The raw log-likelihood ratio is defined as log P(ALT) − log P(REF). For Mendelian and SGE evaluation, deleteriousness is oriented as the negative of the mean forward- and reverse-complement log-likelihood ratio, so larger scores indicate a larger likelihood penalty for the alternate allele. Forward and reverse-complement scores are retained separately, but their mean is the default MarinDNA and Evo 2 score used in the headline comparison. GPN-Star is represented by its calibrated likelihood-ratio score, whereas AlphaGenome is represented by its maximum L2 reference/alternate prediction score; these scores are not interchangeable measurements of model likelihood.

4.8. Frozen-embedding linear probes

The same forward and reverse-complement passes used for likelihood scoring also produce last-layer hidden states for the reference and alternate alleles. Hidden states are mean-pooled across the entire DNA window with special tokens excluded, accumulated and strand-averaged in float32, and stored in float16. For directional Mendelian and SGE labels, the probe feature concatenates the reference embedding with the alternate-minus-reference embedding after upcasting to float32. A separate StandardScaler plus L2-regularized logistic-regression pipeline is fit within each consequence subset. Predictions use leave-one-chromosome-out cross-validation, with the regularization strength retuned inside each outer fold by chromosome-grouped cross-validation over a fixed 17-point grid from 1e-12 to 1e4. Probe and zero-shot metrics are calculated on identical variant rows.

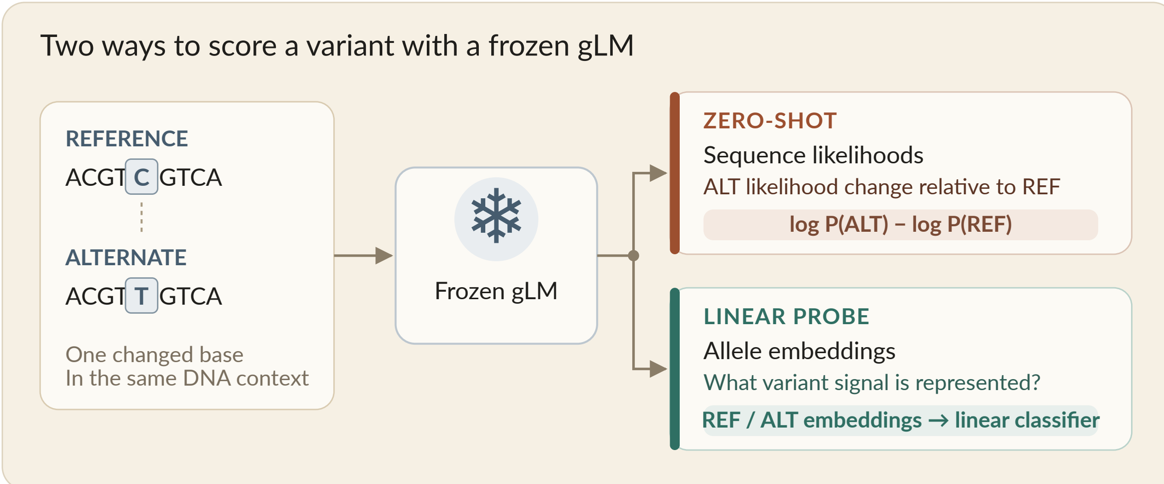

Figure 18 **Variant-effect prediction scoring approaches.** The zero-shot likelihood score and frozen-embedding linear probe use the same reference and alternate sequence inputs.

Figure 18 contrasts the two readouts applied to every frozen checkpoint.

4.9. Metrics and uncertainty

Performance is summarized by area under the precision–recall curve (AUPRC). For matched Mendelian zero-shot analyses, AUPRC is computed within consequence subset and uncertainty is estimated by resampling matched groups with replacement, thereby preserving the one-to-many matched structure. For probe analyses, AUPRC is first calculated per chromosome and then weighted by the number of contributing variants; uncertainty is estimated by a chromosome-cluster bootstrap. SGE AUPRC is computed separately within each MaveDB accession and consequence subset before macro-averaging because scores are not comparable across studies. Unless a caption states otherwise, error bars denote ±1 bootstrap standard error. Paired model comparisons on matched variants resample matched groups once per iteration and recompute both AUPRC values on the same resample. The comparison reports the observed AUPRC difference, its bootstrap standard error, the 2.5th and 97.5th percentiles of the paired difference distribution, and a two-sided bootstrap p-value. Comparative language in the title, abstract, Results, and Discussion follows this paired uncertainty rather than the ordering of point estimates alone.

4.10. Training compute and inference throughput

Training compute is reported as model-training FLOPs for the complete runs. The final m5.1 model used approximately 1.1e21 FLOPs over 166B tokens, whereas Evo 2 40B reports approximately 2.25e24 FLOPs over 9.3T tokens. Inference throughput was measured on one NVIDIA GH200 for steady-state scoring with forward- and reverse-complement passes and embedding output enabled. The workload scores one million variants at each model's native context length: 256 tokens for m5.1 and 8,192 tokens for Evo 2 40B. The resulting comparison is an as-deployed workload measurement rather than a same-context, same-batch, or per-token architecture benchmark.

5. Data, Code, Models, and Live Resources

The MarinDNA repository contains the data pipelines, evaluation implementation, manuscript source, and tracked investigation scripts. The mechanical manuscript and frozen figure baseline is available at commit 3b608d39b41c2330636ec647dbb25d26b0895187. The final preprint revision will identify commit-pinned training, dataset-construction, evaluation, and plotting code for each retained result.

The frozen Mendelian benchmark is bolinas-dna/evals_mendelian_traits at revision 4aed58e. The frozen saturation-genome-editing benchmark is bolinas-dna/evals_sge at revision 225d3d1. Training datasets, evaluation datasets, and the released model are grouped in the MarinDNA Hugging Face collection. The exact public model and training-dataset revisions will be pinned here before the preprint is released.

The MarinDNA leaderboard is an evolving project resource. Its contents may change as models and baselines are added and must not be treated as the frozen comparison reported in this manuscript. The committed manuscript figures and their recorded input revisions define the archival result snapshot.

Optional interactive resources include:

- Interactive sequence explorer.
- Model inference and BRCA1 variant-effect prediction notebook.

6. Open Development and Provenance

MarinDNA was developed through public, issue-tracked experiments in which hypotheses, configurations, intermediate results, and negative findings were recorded as the work progressed. The issue history provides development provenance but is not required scientific exposition: the manuscript states the methods, evidence, and limitations needed to interpret its claims. Decisions and substantial editorial changes for this frozen preprint are tracked in issue #449.

Key result families retain the following development records, which will be expanded with exact comments, code commits, checkpoints, dataset revisions, and figure inputs in the supplementary provenance table:

- Data specialists and mixture balance: #21, #27, and #13.
- Manual scaling failure and the need for transfer: #57.
- Validation-loss design and interpretation: #8.
- Frozen-embedding probe development: #369 and the evaluation-pipeline records it references.
- Throughput measurement: #354.

Issue links are provided to make the branching research history inspectable; scholarly claims about prior work use bibliographic citations, and claims about MarinDNA are supported by the manuscript's figures, Methods, and frozen artifacts.

7. Acknowledgements

We thank Isaac Hodes, Yael Elmatad, David Hall, Will Held, and Jeff Hammerbacher, as well as others across the Open Athena and Marin communities, for thoughtful discussions. We also thank the Google TPU Research Cloud (TRC) program for providing compute resources.

8. Author Contributions

TODO: Complete after the author list and order are agreed.

9. Competing Interests

TODO: Obtain and record declarations from all authors before public posting.

10. Funding and Compute Resources

Training used compute resources provided by the Google TPU Research Cloud program. Additional funding and resource disclosures will be completed before public posting.

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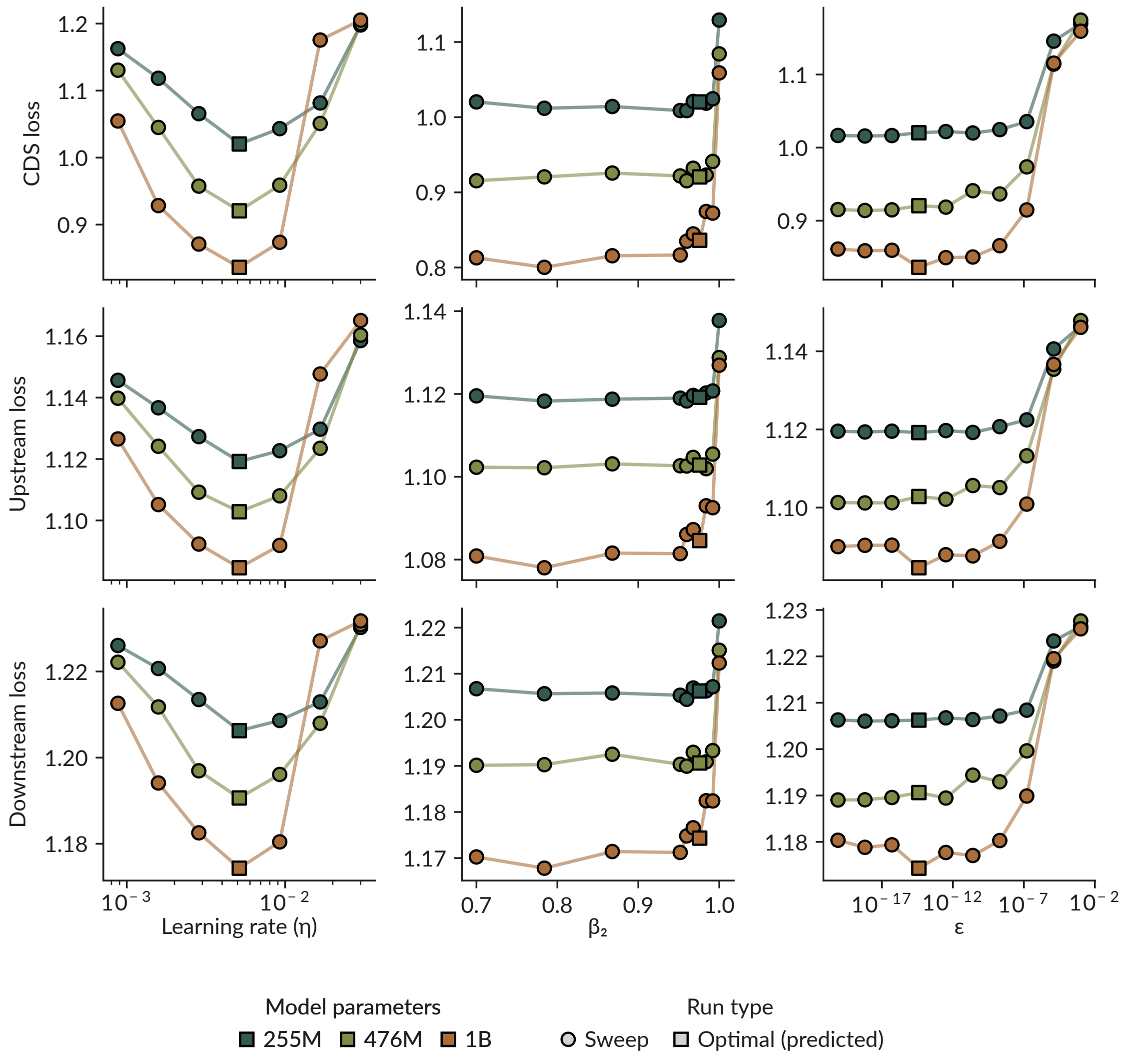
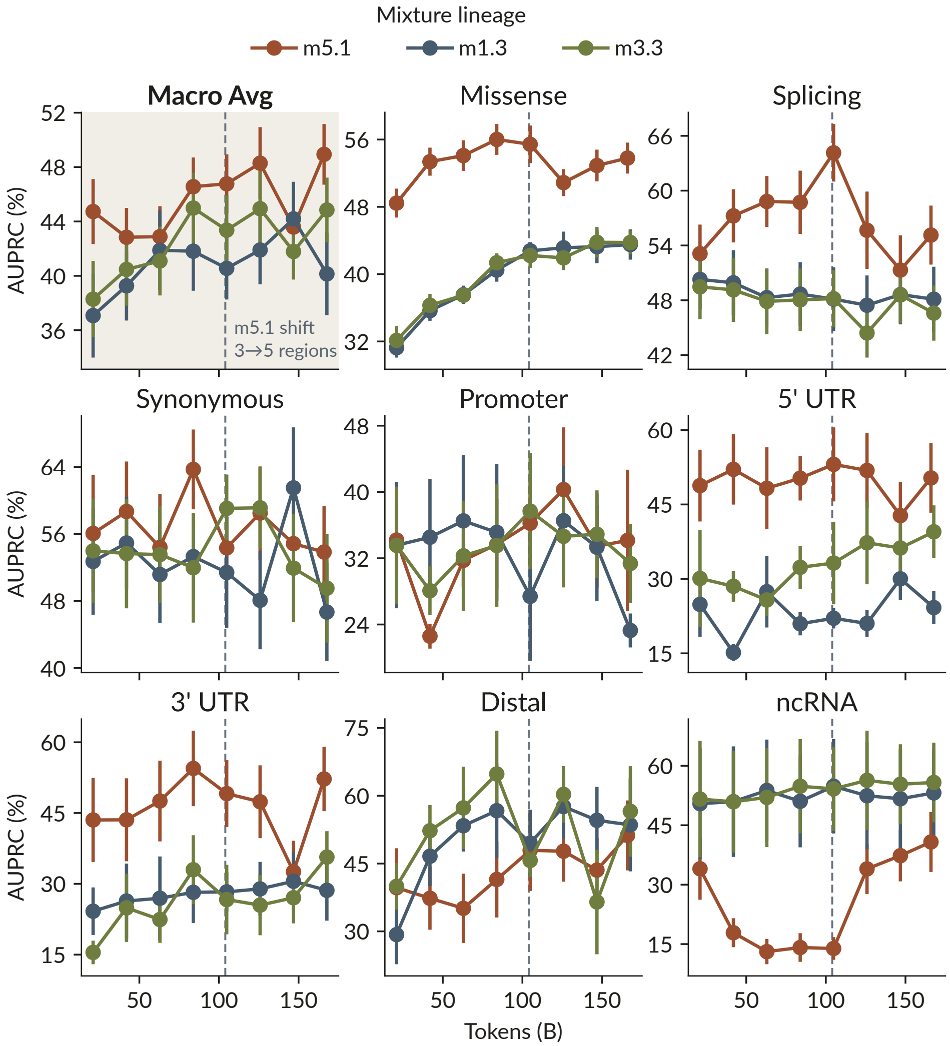
Supplementary Information

The figures below were hidden behind disclosure controls in the published blog presentation and are included here as supplementary figures.



Supplementary Figure S1 **Region-specific hyperparameter transfer.** Validation loss as a function of learning rate, β_2, and ε, evaluated separately for CDS, upstream, and downstream regions.



Supplementary Figure S2 **Linear-probe mixture-lineage trajectories.** Linear-probe Mendelian AUPRC (chromosome-averaged) versus training tokens for three model-mixture lineages. Error bars denote SE.